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Internal anatomy of the brainstem I: general layout and major nuclei

Isaac Kurtzer, Ph.D.
Department of Biomedical Science
ikurtzer@nyit.edu

Session objectives

- 1) Students will be able to understand the divisions of the medulla, pons, and midbrain from external views and transverse sections.
- 2) Students will be able to identify the major structures, reflecting nuclei and tracts, on the external anatomy of the brainstem, along with cranial nerves and ventricles.
- 3) Students will be able to identify the major nuclei on transverse sections of the brainstem and cerebellum.
- 4) Students will be able to describe the placement of the cranial nerves on the brainstem.

The Human Brain: An introduction to its functional anatomy by J. Nolte
Chapter 11 “Organization of the brainstem”

<http://arktos.nyit.edu/login?url=https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323653985000114>

The brainstem involves many systems & functions

- Cranial nerves and cranial nerve nuclei
- Assorted major nuclei (e.g. substantia nigra)
- Reticular formation
- Ascending & descending pathways (e.g. corticospinal tract)

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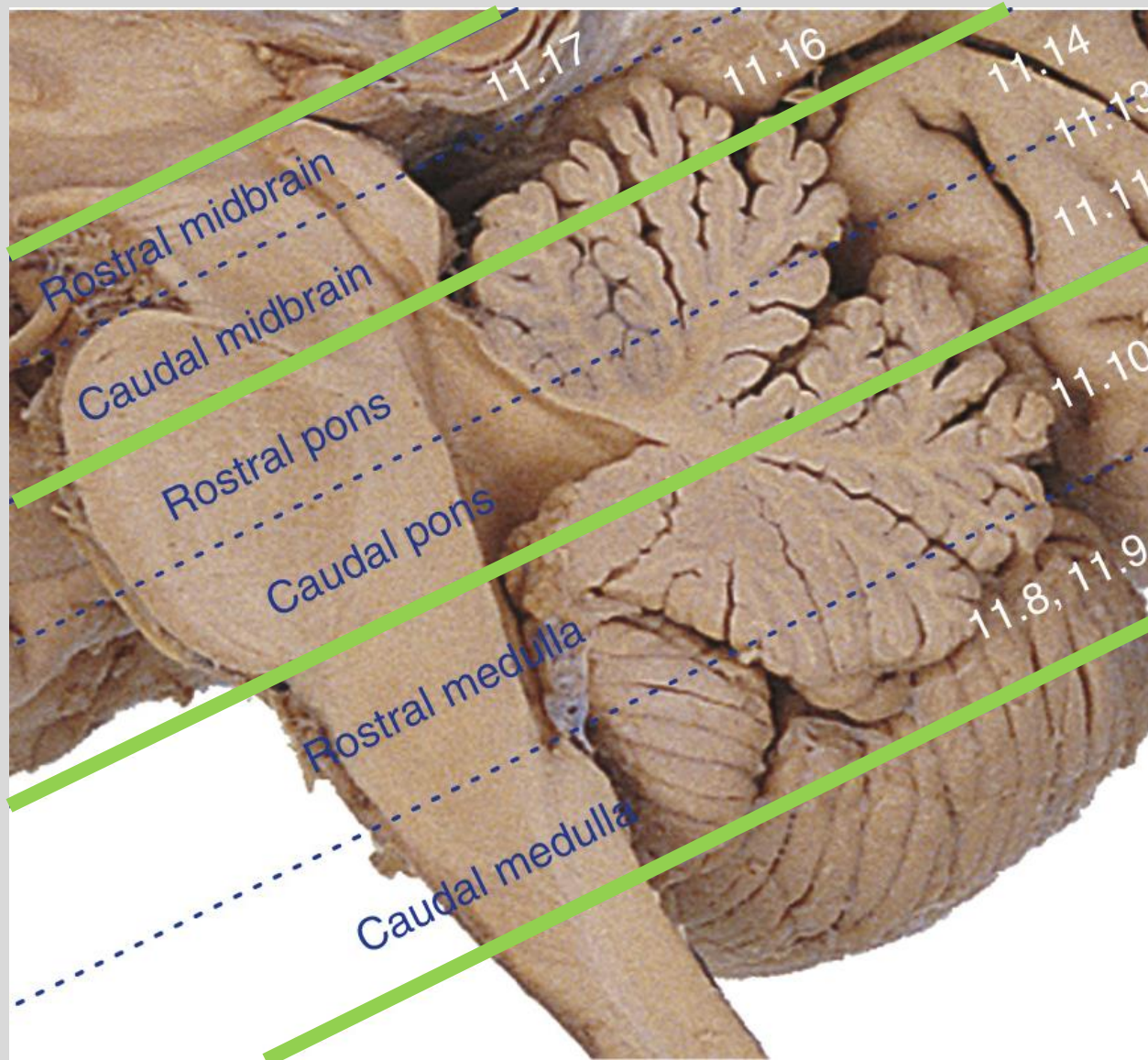


The largest divisions of the brainstem: medulla, pons, and midbrain

Midbrain

Pons

Medulla



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First identity global region,
then narrow down to
specifics.

Important differences within
the rostral-caudal extent of
the major divisions

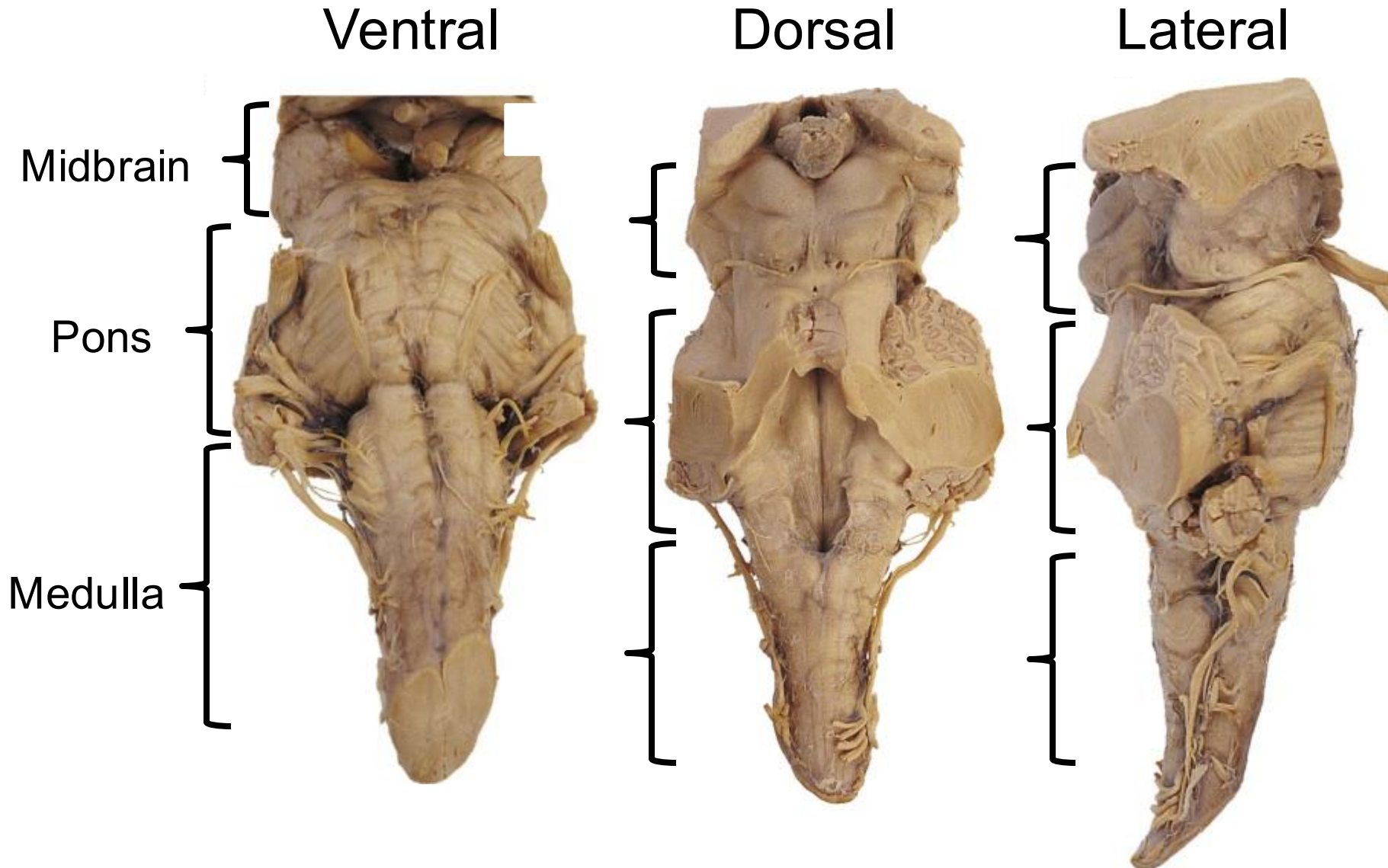
The cerebellum and vesicles
are complementary to the
brainstem.

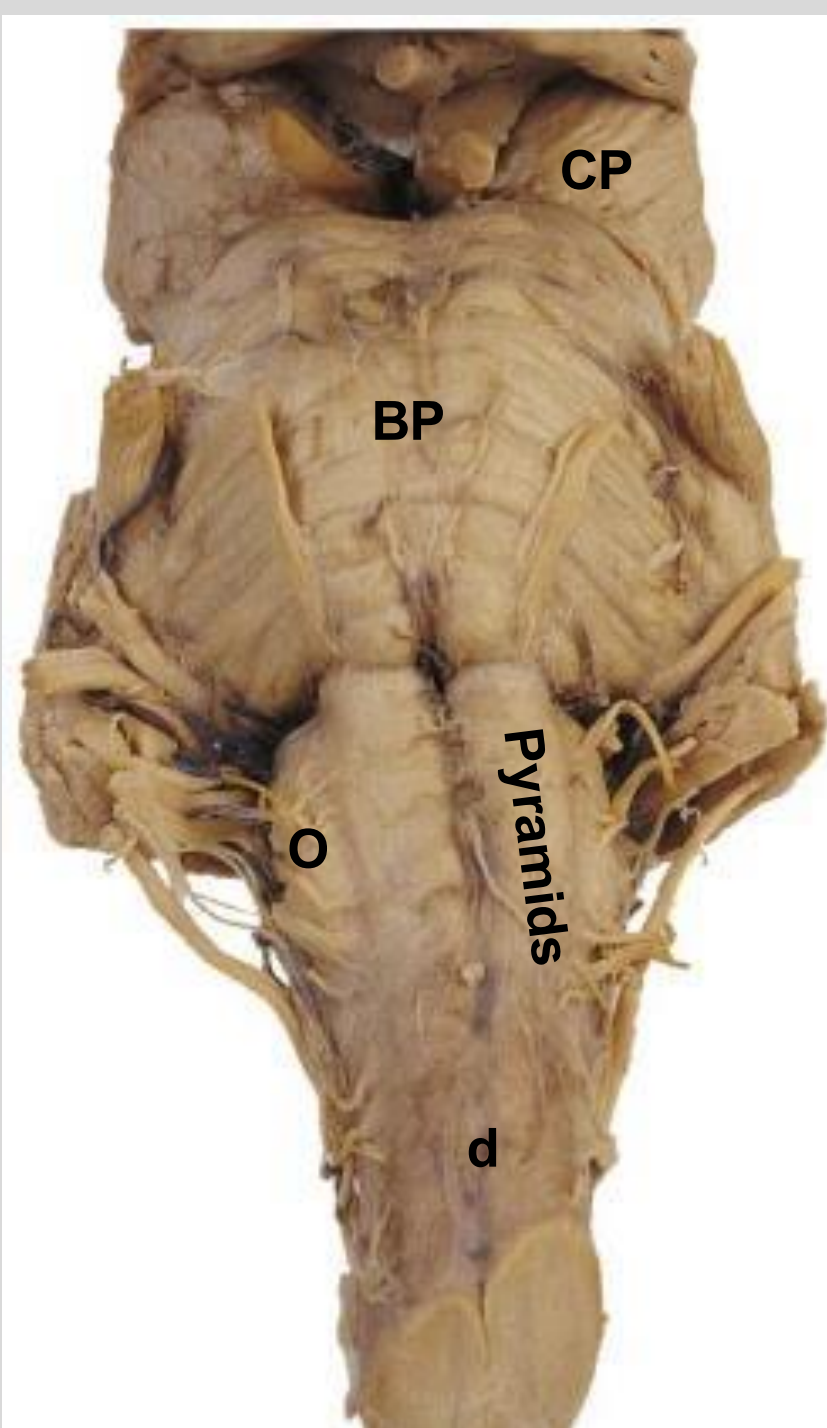
External views of the tracts & nuclei of the brainstem

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Ask yourself:
“What external features
reflect a structure
underneath?”





Cerebral peduncle – thick columns
Anterior portion = crus cerebri which
includes descending tracts from
cerebral cortex.

Basilar pons (BP) – bulb-shaped

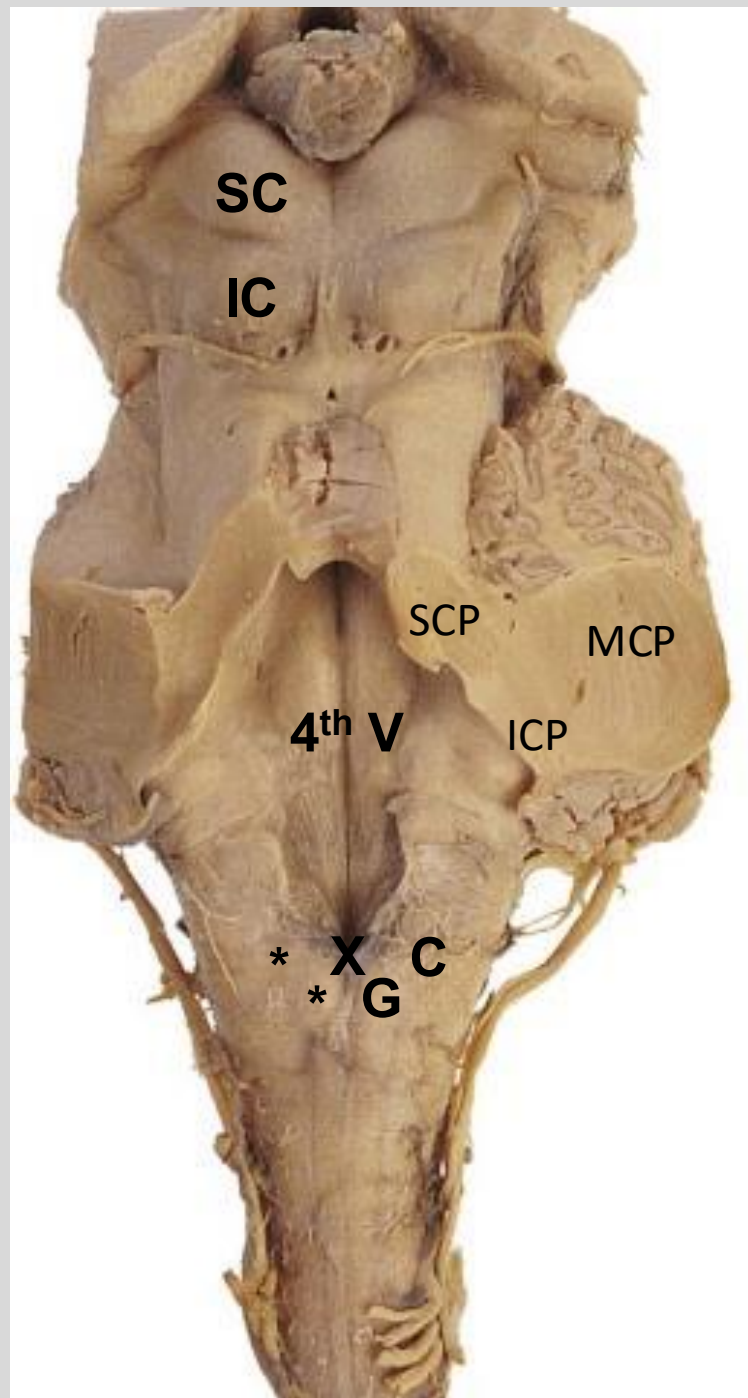
Medulla – spinal cord-like
Medullary pyramids (inside is the corticospinal tract)
Pyramidal decussation (d)
Olive (O) (inside is the inferior olivary nucleus)

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Ventral brainstem

Dorsal brainstem

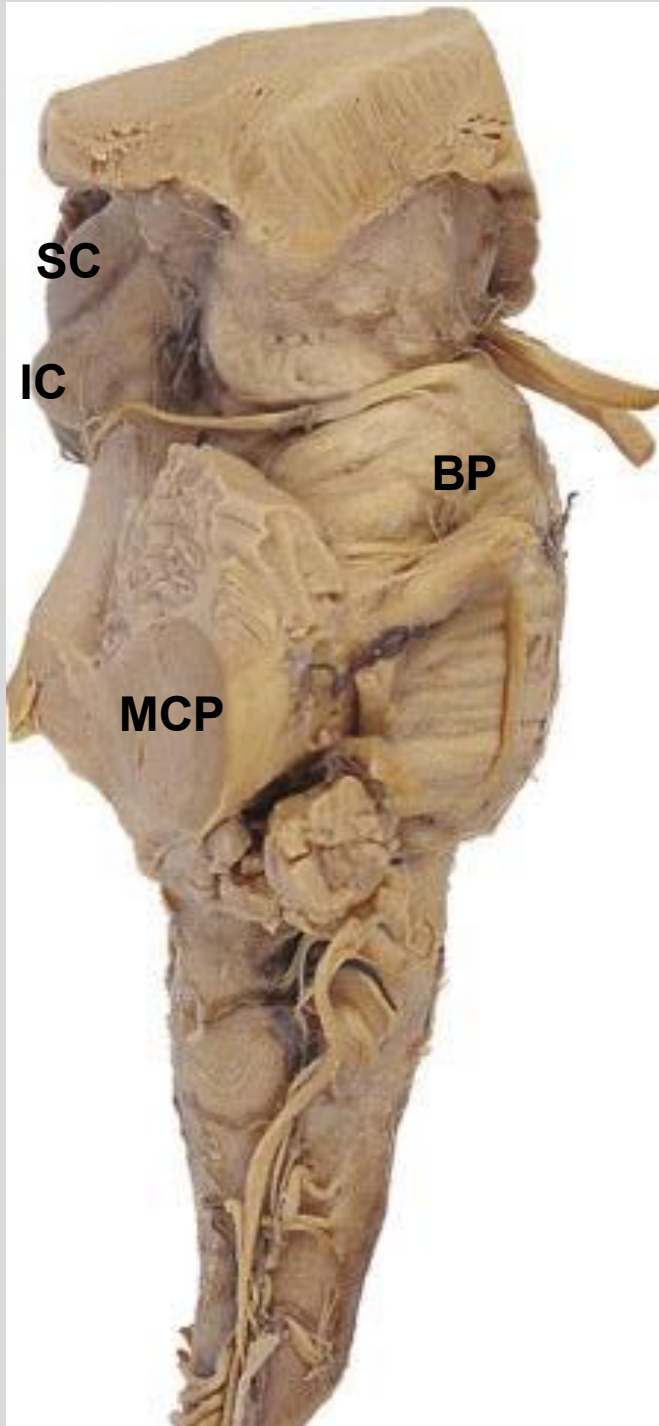


Superior colliculus (SC)] tectum
Inferior colliculus (IC)

Superior cerebellar peduncle (SCP)
Middle cerebellar peduncle (MCP)
Inferior cerebellar peduncle (ICP)

Floor of 4th ventricle (4th V)

Cuneate tubercle (C) (inside is the cuneate nucleus)
Gracile tubercle (G) (inside is the cuneate nucleus)
Obex (X)



Superior colliculus (SC)
Inferior colliculus (IC)

Basilar pons (BP)
Middle cerebellar peduncle (MCP)

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**Lateral brainstem
(dorso-lateral view)**

Stained sections of the brainstem

nuclei: light area in myelin-stained slides

axons: dark area in myelin-stained slides

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Anterior-posterior regions

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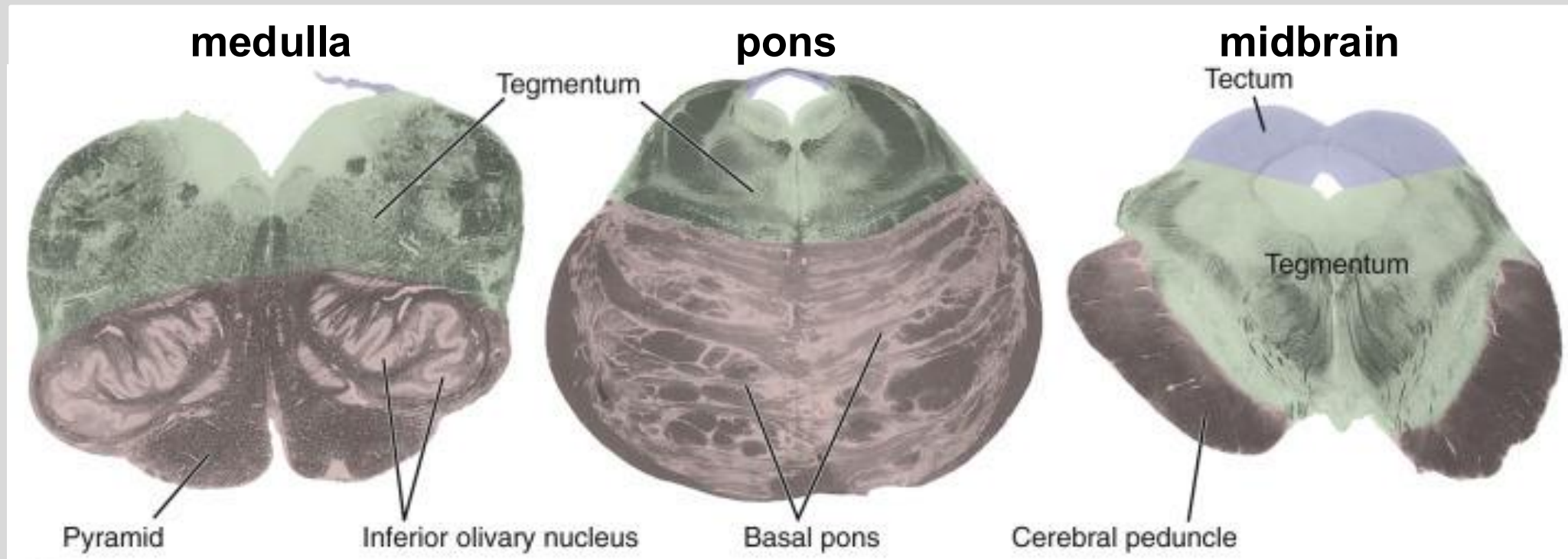
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Tectum is the superior and inferior colliculi

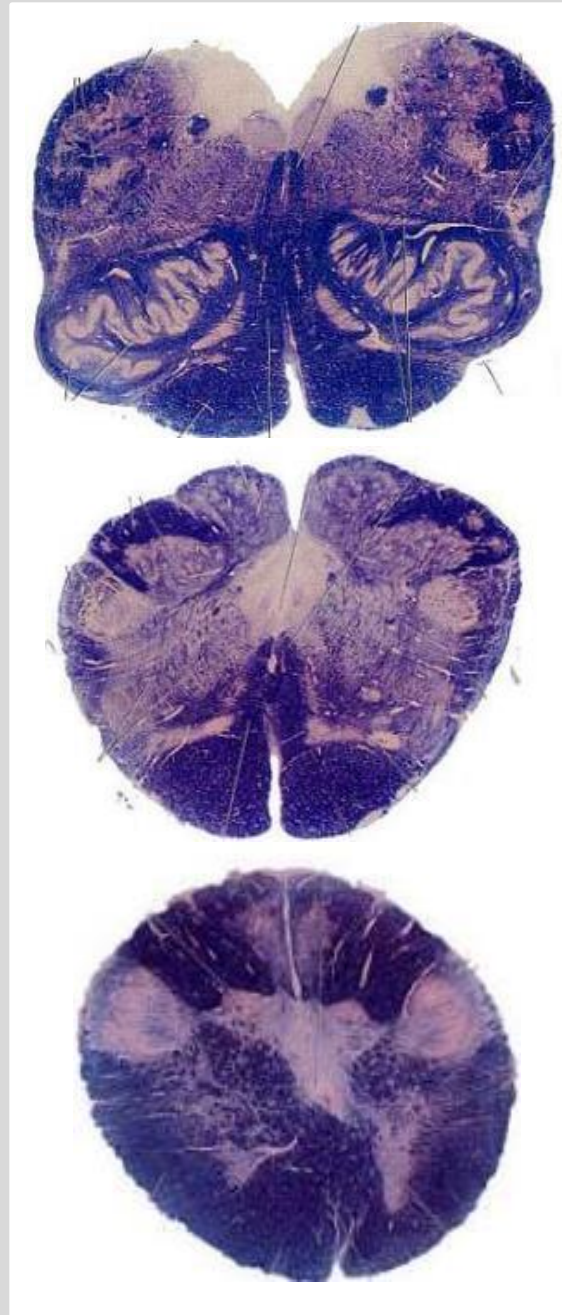
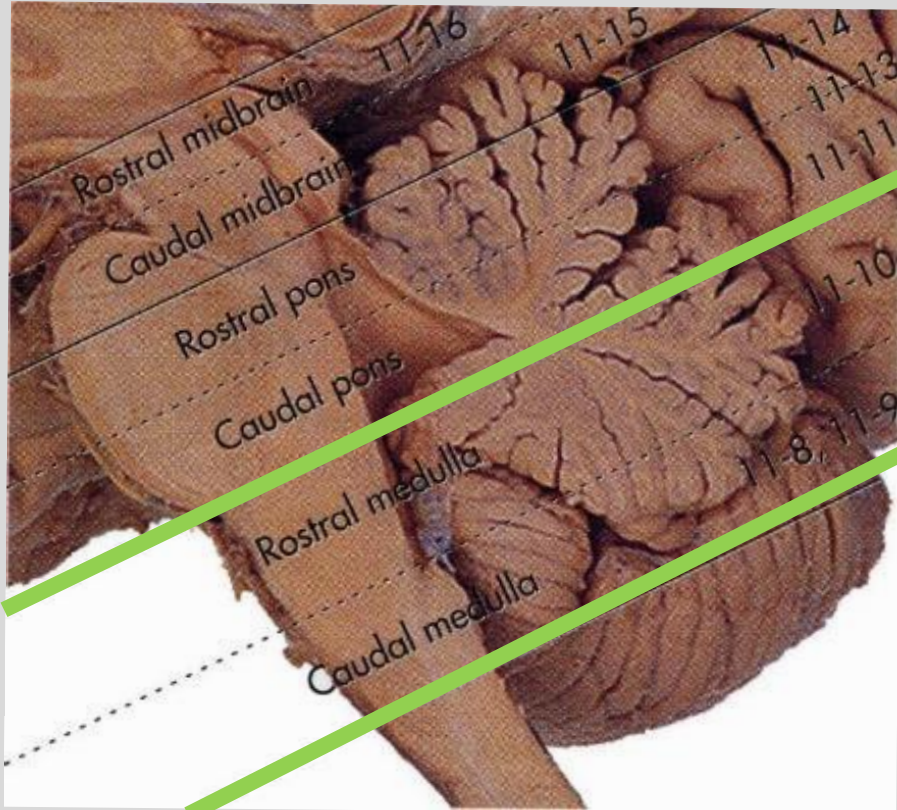
Tegmentum is the region surrounding (and largely anterior) to the 4th ventricle & cerebral aqueduct.

It includes cranial nerve nuclei, reticular formation and some assorted tracts.

Basilar region is anterior to the tegmentum and includes major descending tracts.



Medulla



rostral

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Overall tubular shape.
Opens at top to the 4th ventricle.

Would inferior olivary nucleus be dorsal or ventral, rostral or caudal?

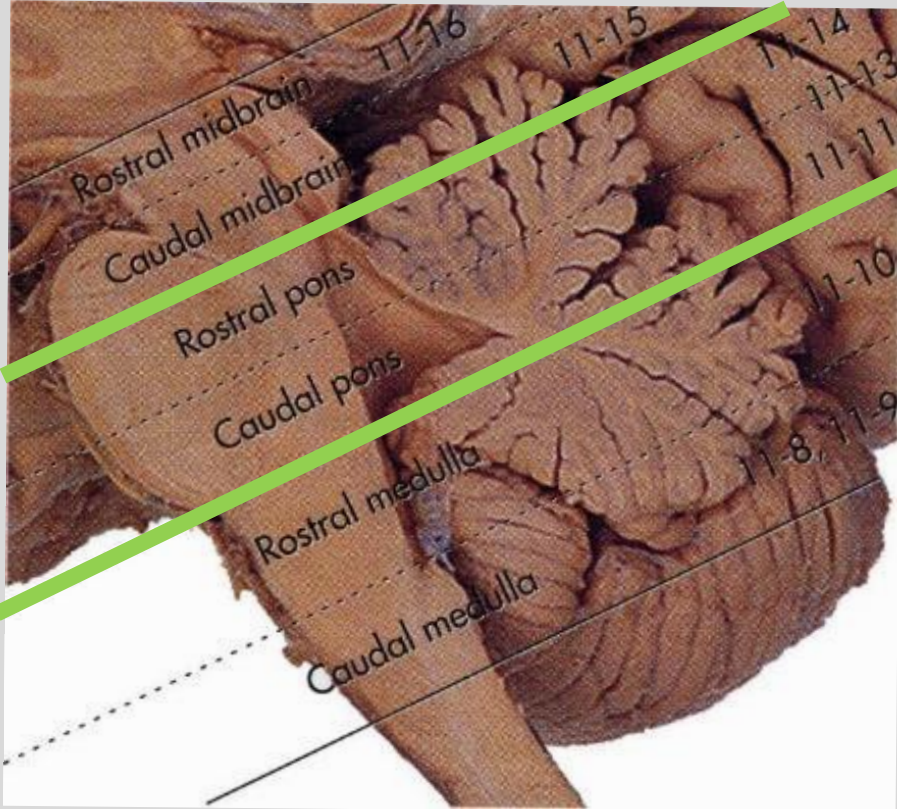
What about gracile and cuneate nuclei?
Hint: also called dorsal column nuclei.

caudal

Pons

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rostral

Very bowed out and “white” throughout the ventral portion

Clear 4th ventricle.

What major brain region is posterior to the 4th ventricle?



caudal

Midbrain

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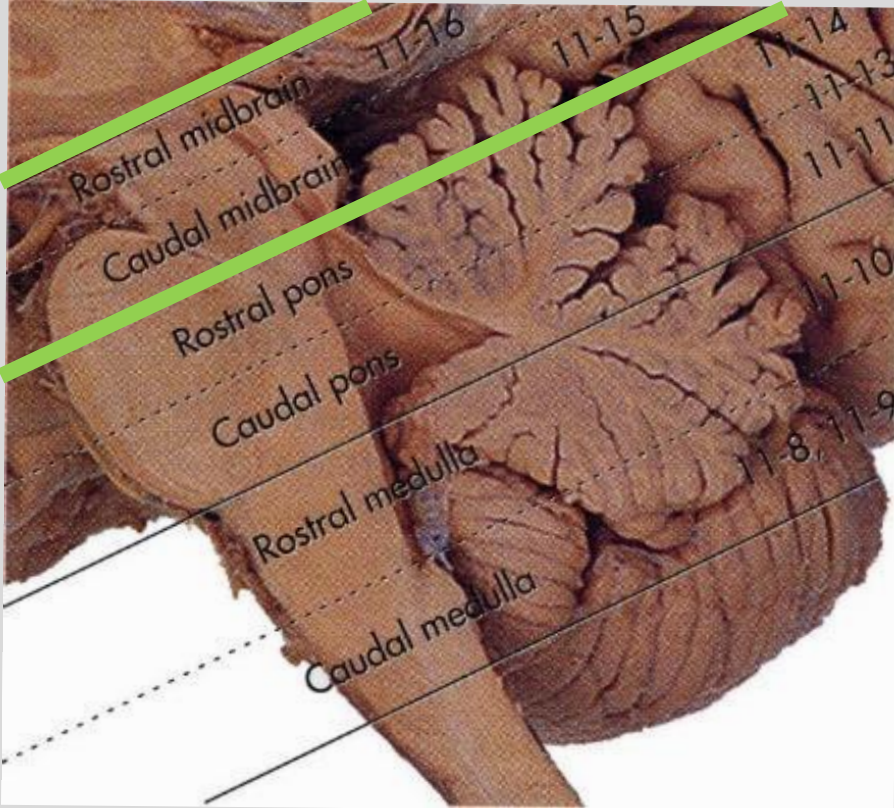
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rostral

Tectum is the superior and inferior colliculi

Tegmentum is the region surrounding (and largely anterior) to the cerebral aqueduct and contains cranial nerve nuclei, reticular formation and some assorted tracts

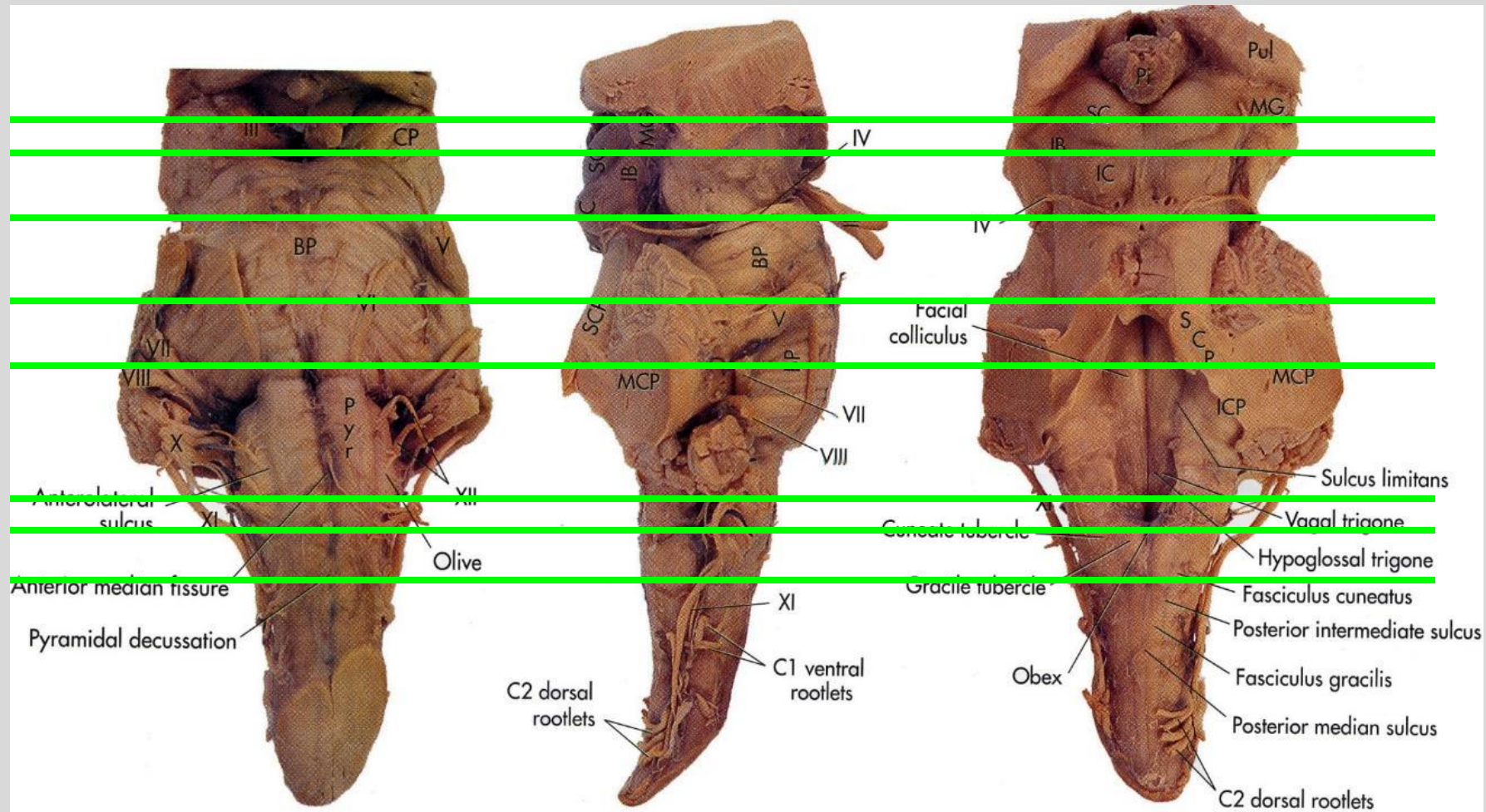
caudal



Where sections align with the external brainstem

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Nuclei to identify (remember their function)

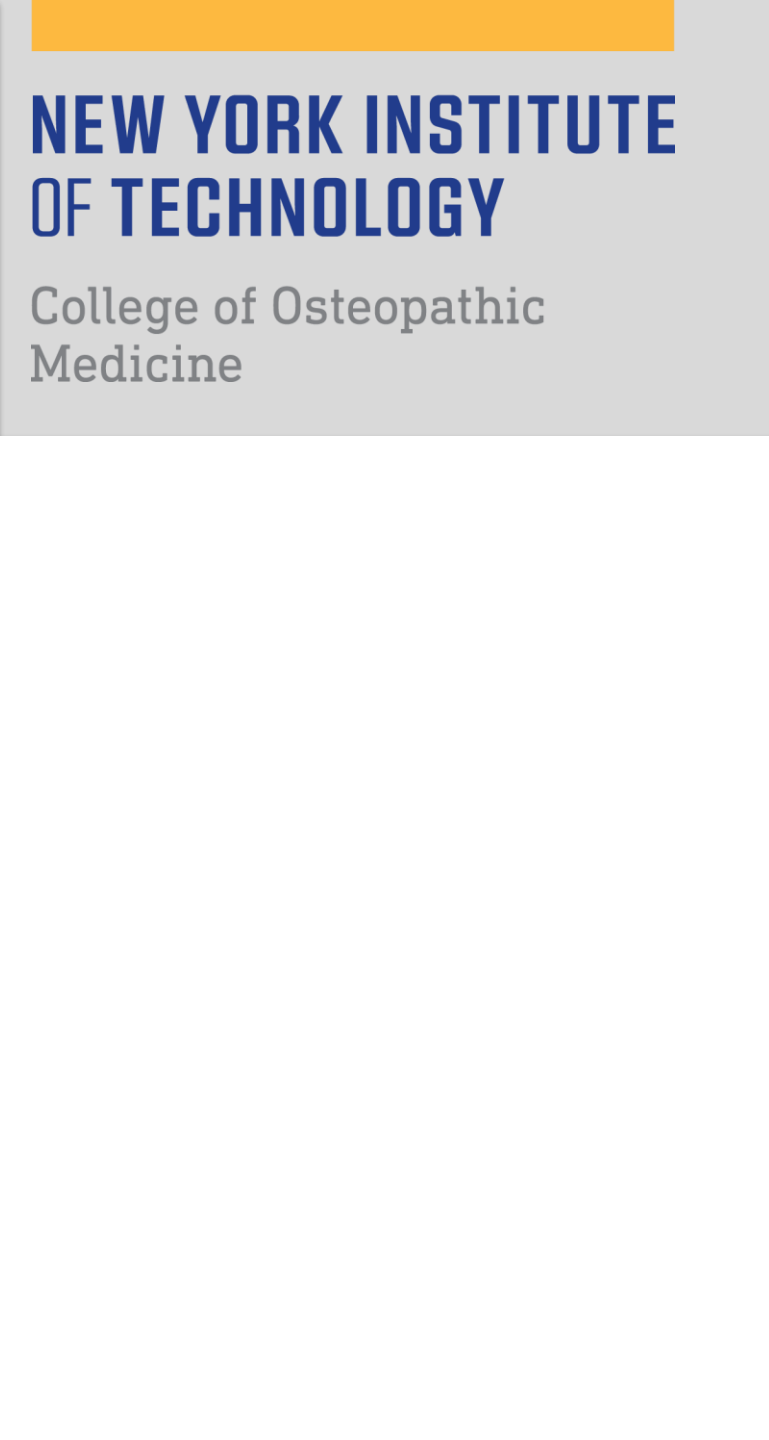
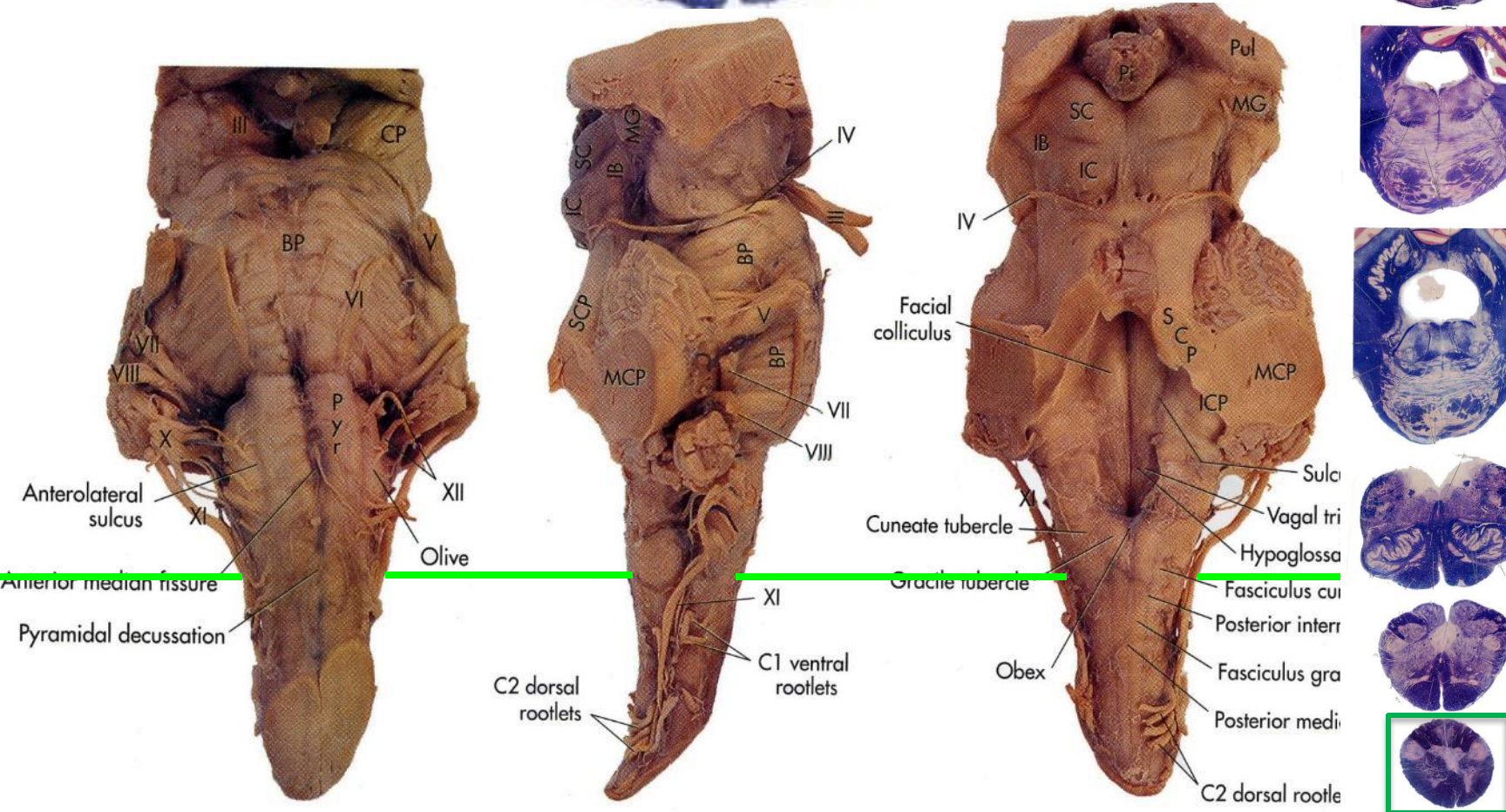
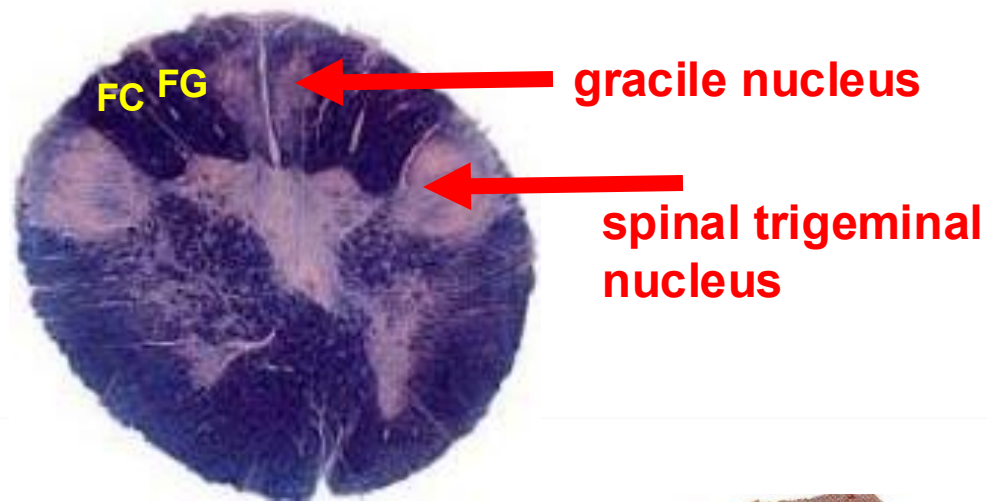
Midbrain	red nucleus	motor-related (contralateral)
	superior colliculus	saccadic eye movements
	inferior colliculus	hearing
	substantia nigra	motor-related (contralateral) <i>linked to basal-ganglia</i>
	periaqueductal gray	pain processing
Pons	pontine nuclei	cortical-cerebellar relay (contralateral)
	dentate nuclei	motor coordination (ipsilateral) <i>pons-level cerebellar nucleus</i>
Medulla	gracile & cuneate nuclei	touch, proprioception (ipsilateral)
	inferior olivary nucleus	motor coordination (contralateral)
	spinal trigeminal nucleus	pain, temperature for face (ipsilateral)

*note that light areas indicate nuclei in myelin-stained slides and this will include cranial nerve nuclei

Medulla

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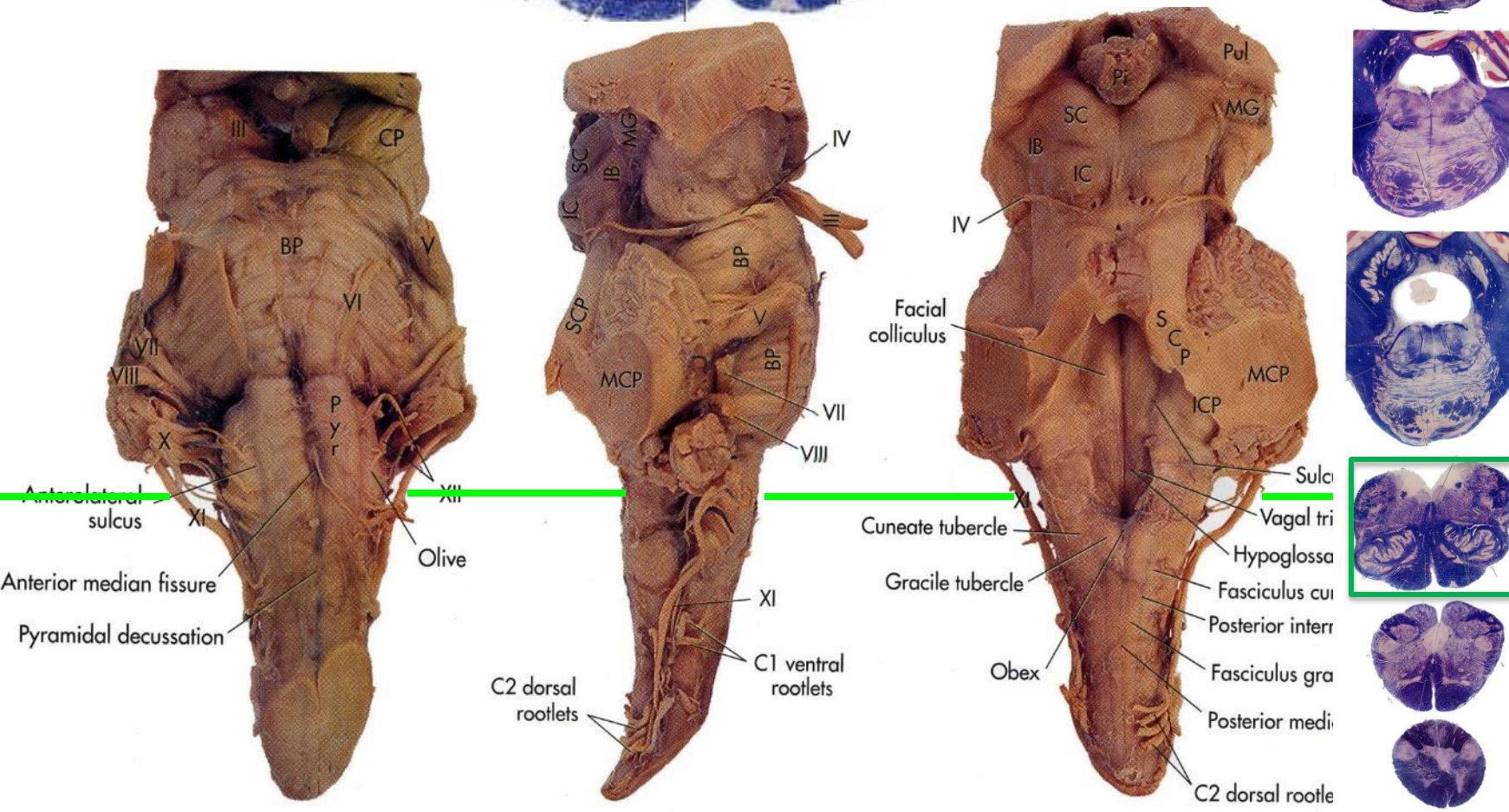
spinal trigeminal nucleus



Medulla

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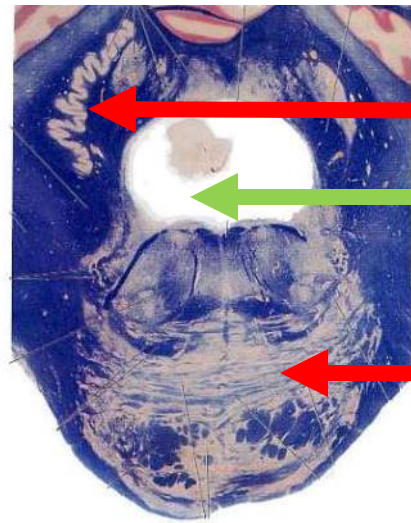
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Pons

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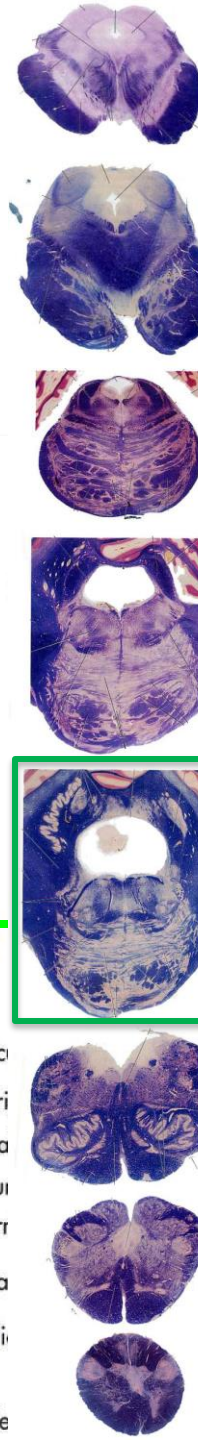
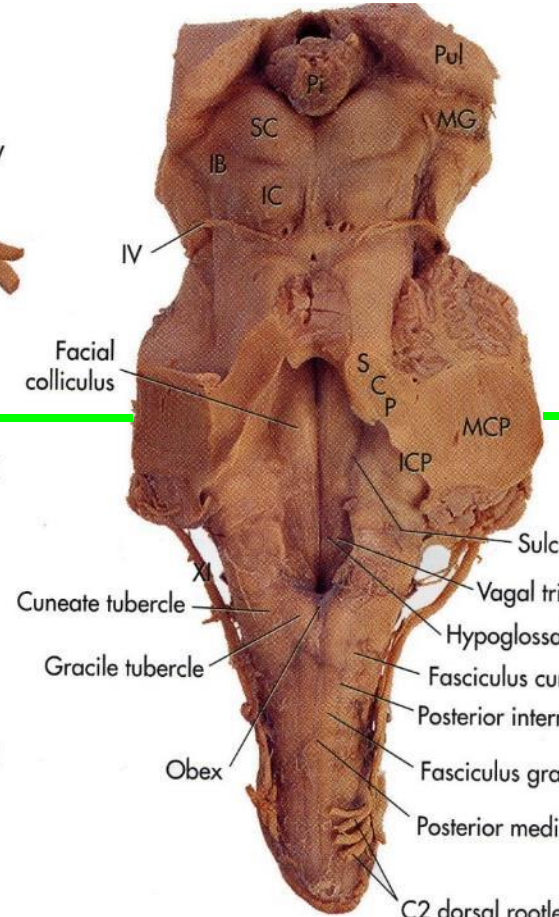
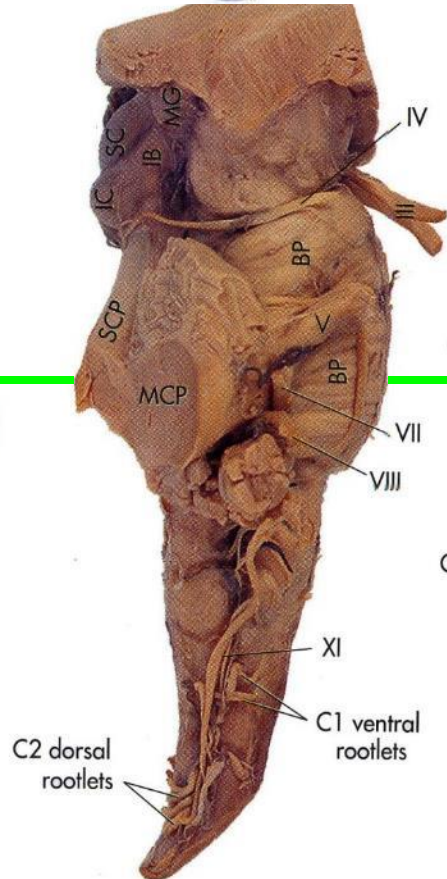
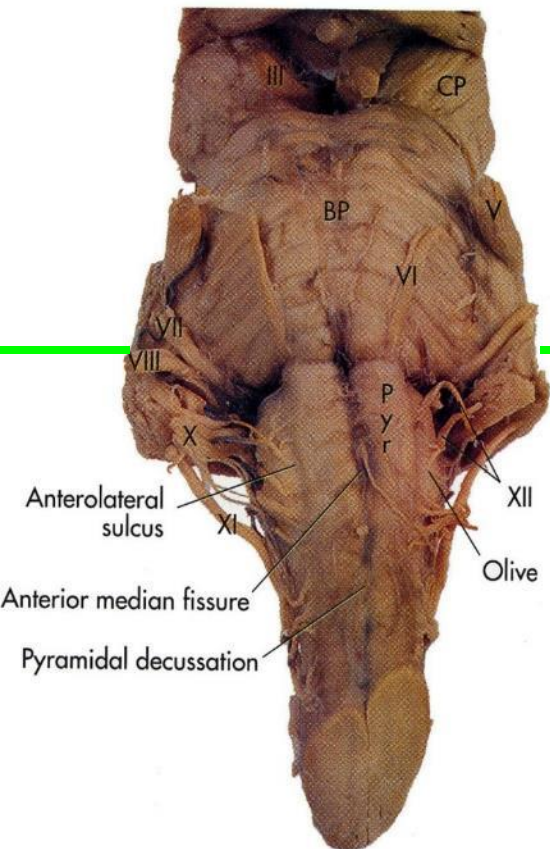
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dentate nucleus

4th ventricle

pontine nuclei



Pons

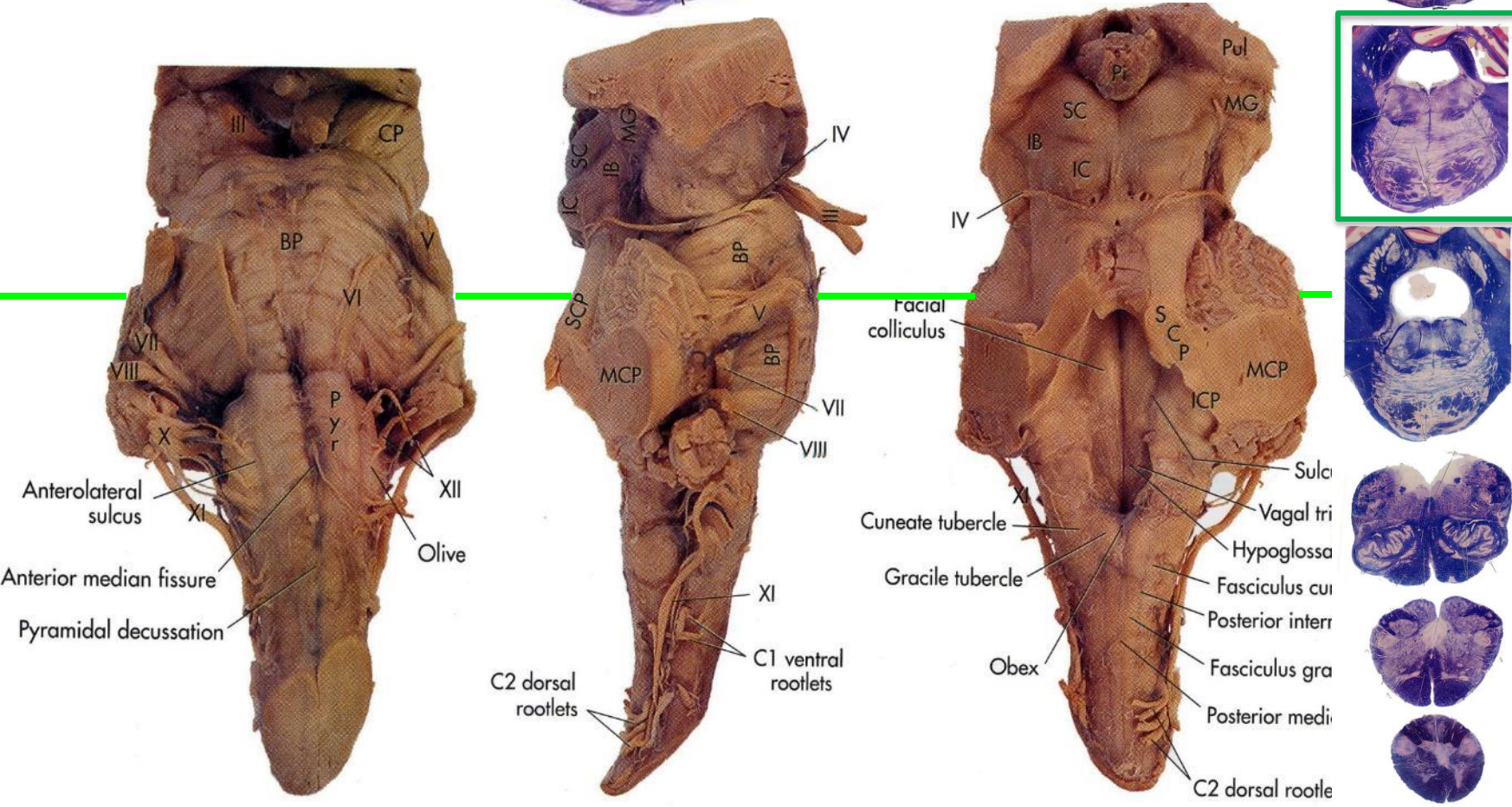
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4th ventricle

pontine nuclei



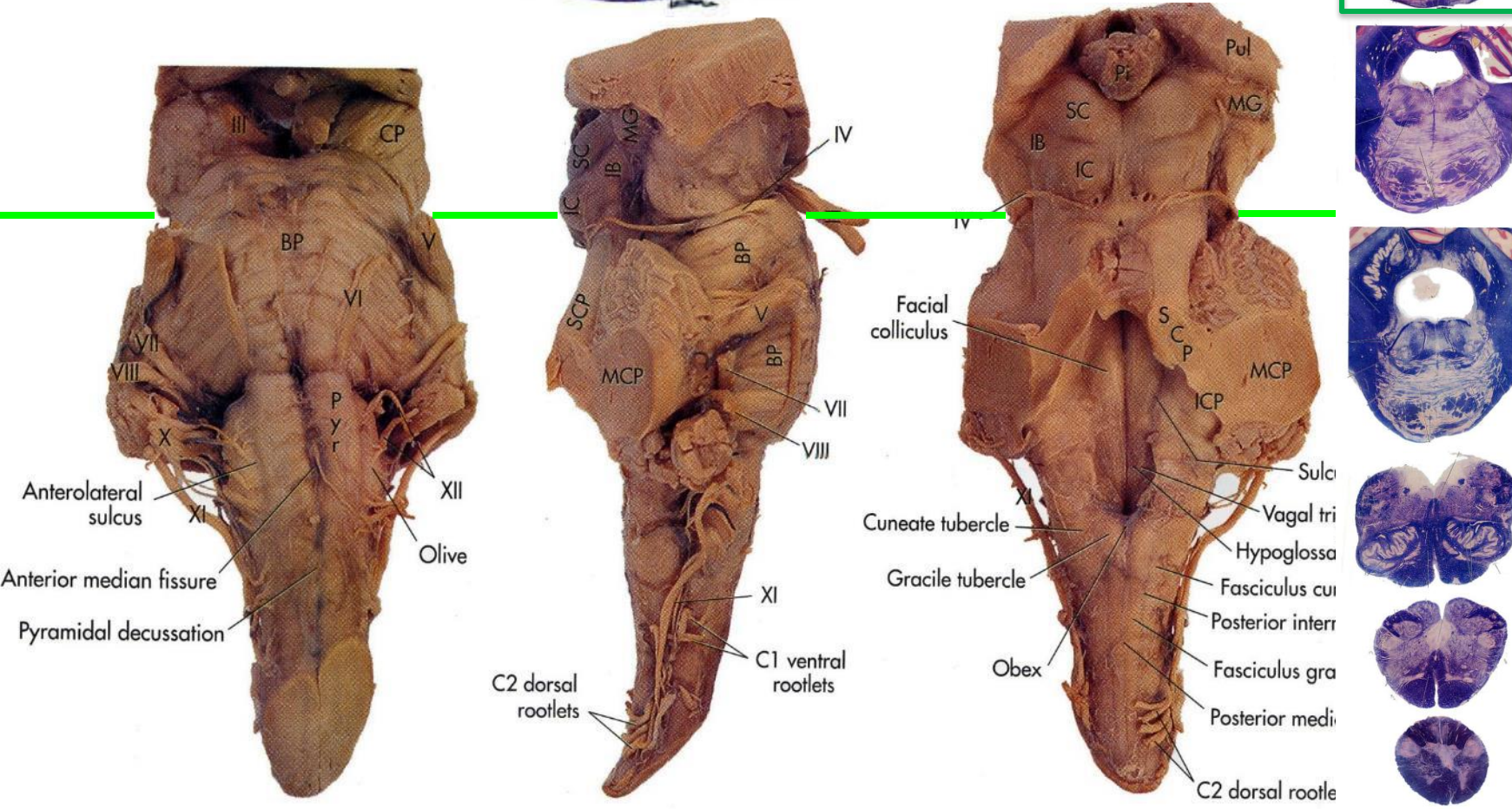
Pons

cerebral
aqueduct/
4th ventricle

pontine nuclei

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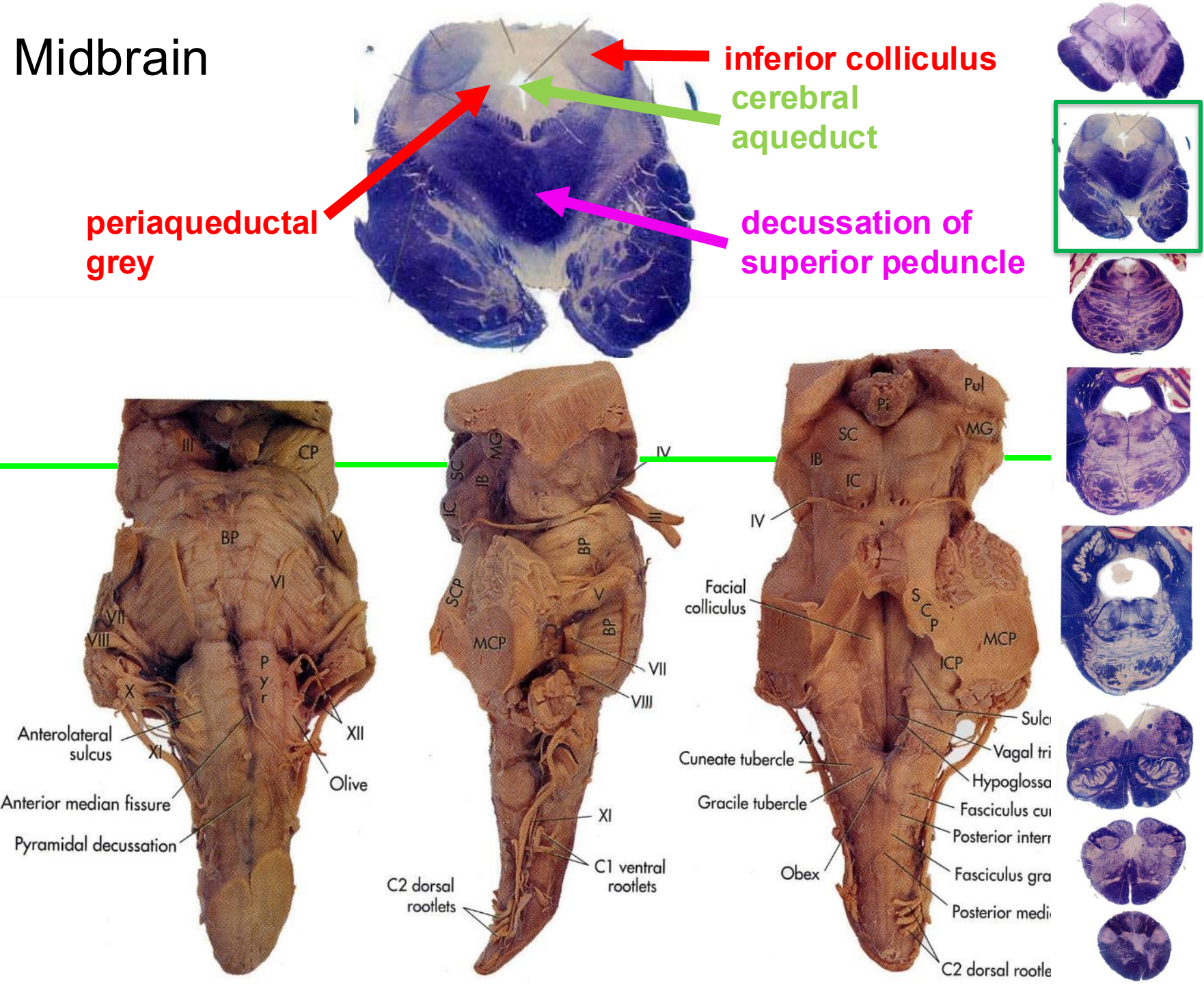
Midbrain

periaqueductal
grey

inferior colliculus

cerebral
aqueduct

decussation of
superior peduncle



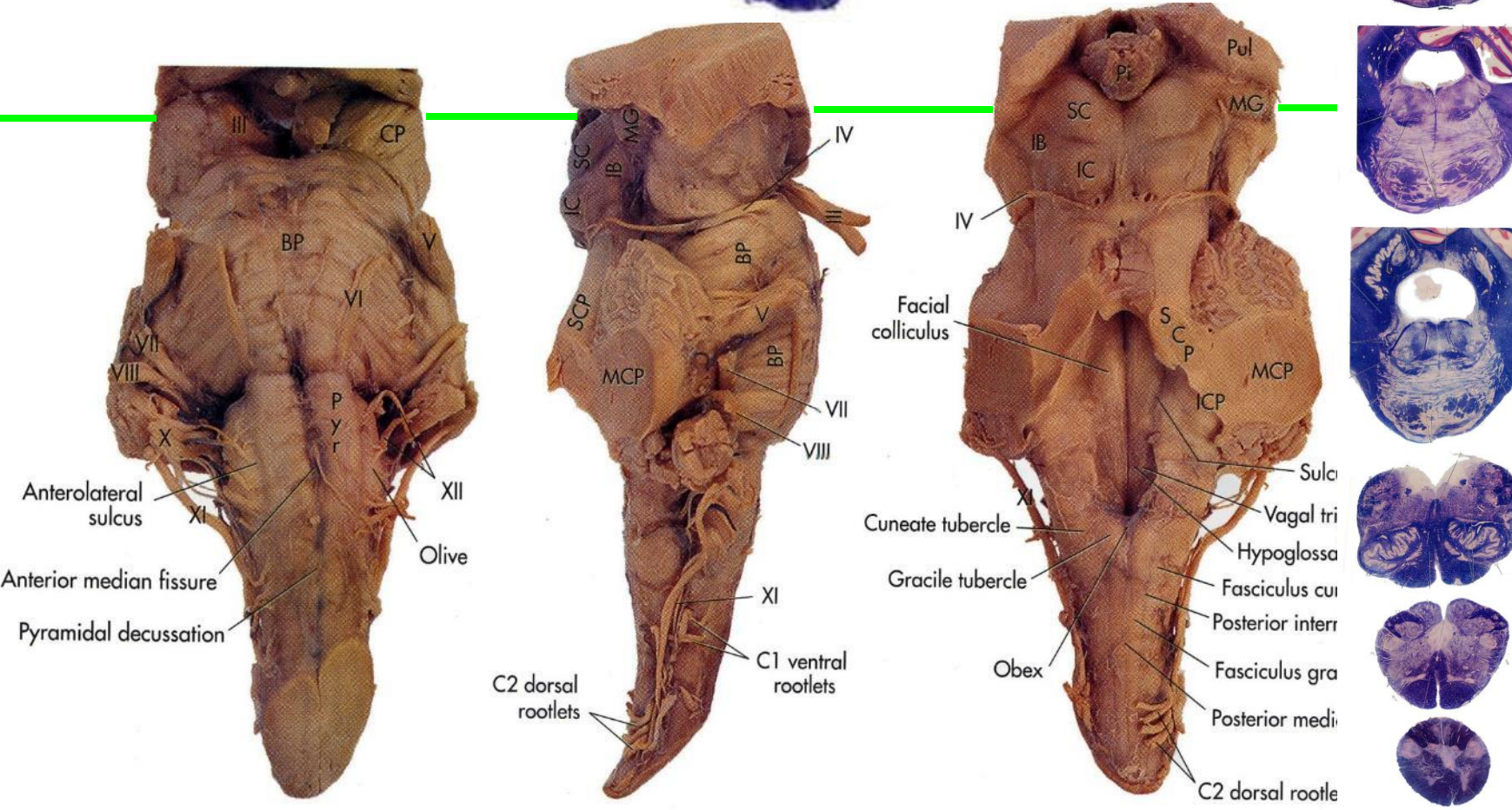
Midbrain

periaqueductal grey

cerebral aqueduct

sup. peduncle efferents

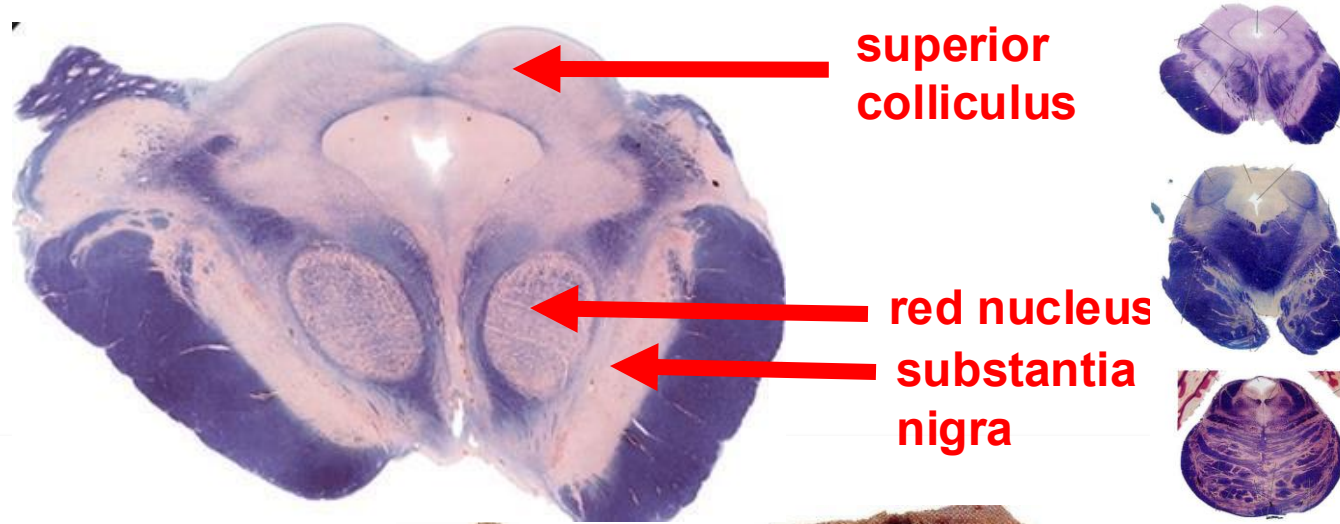
substantia nigra



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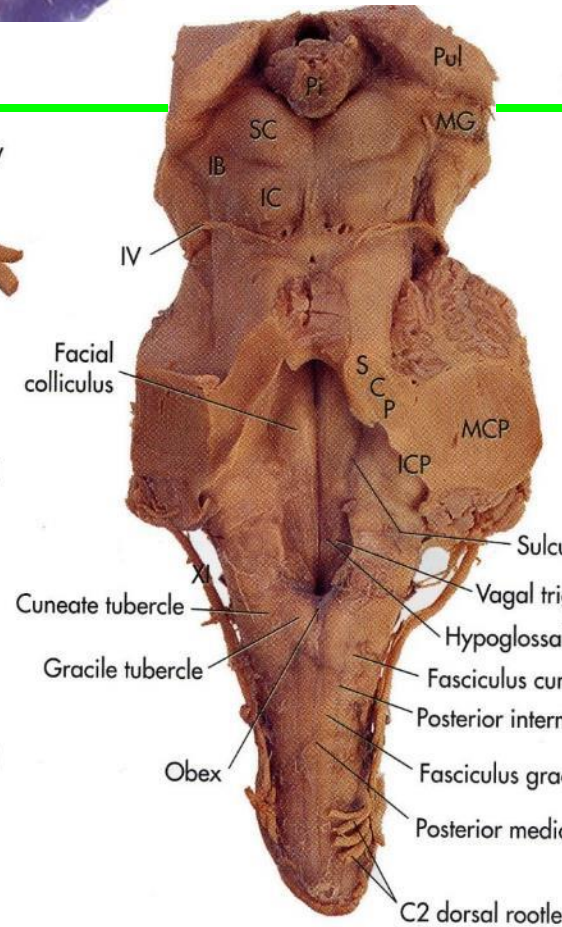
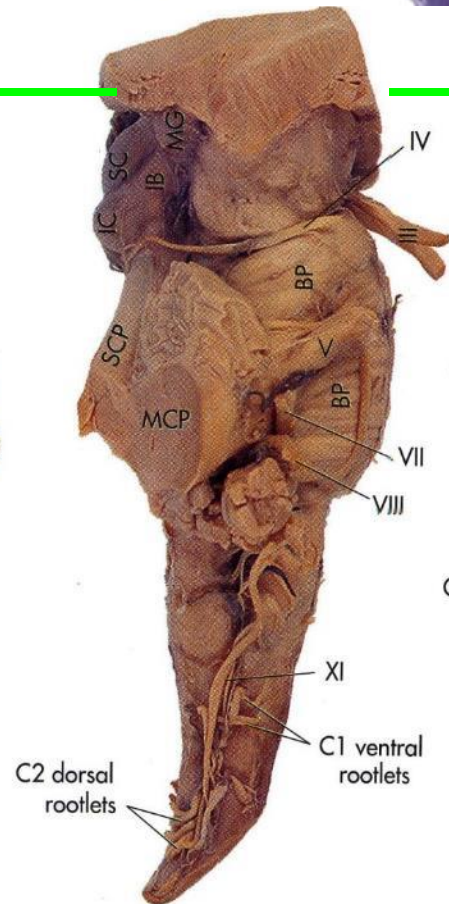
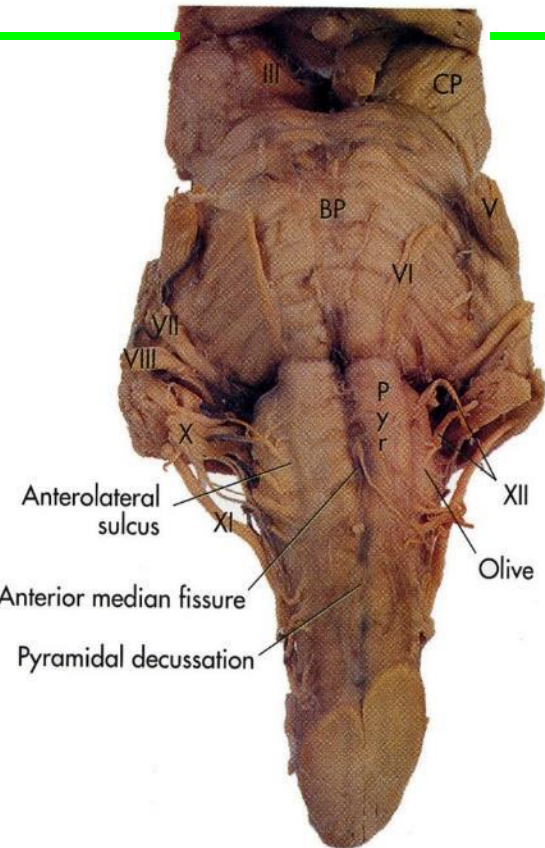
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Midbrain



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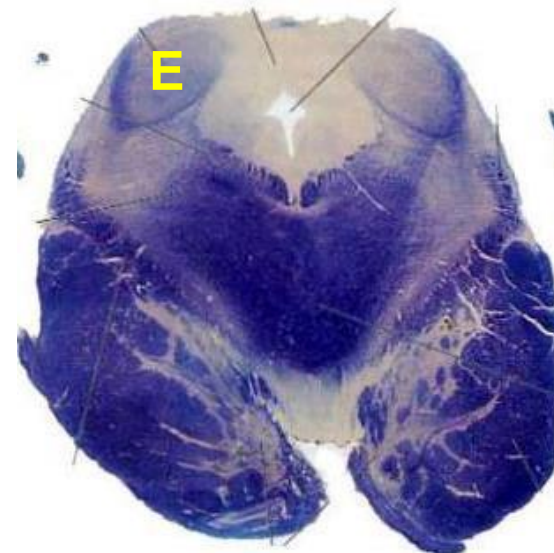
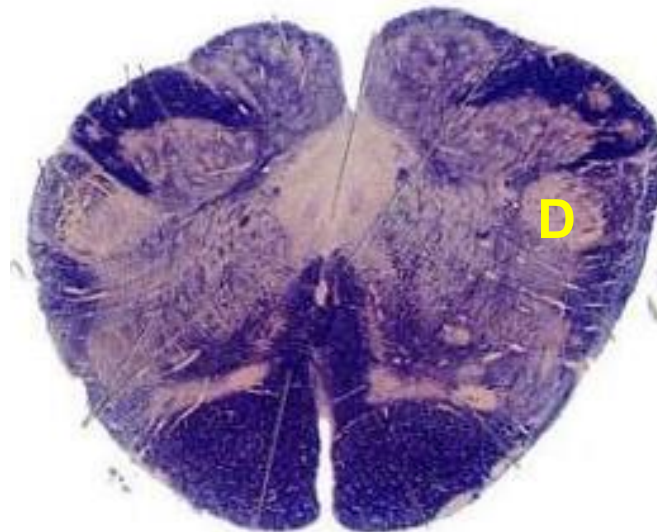
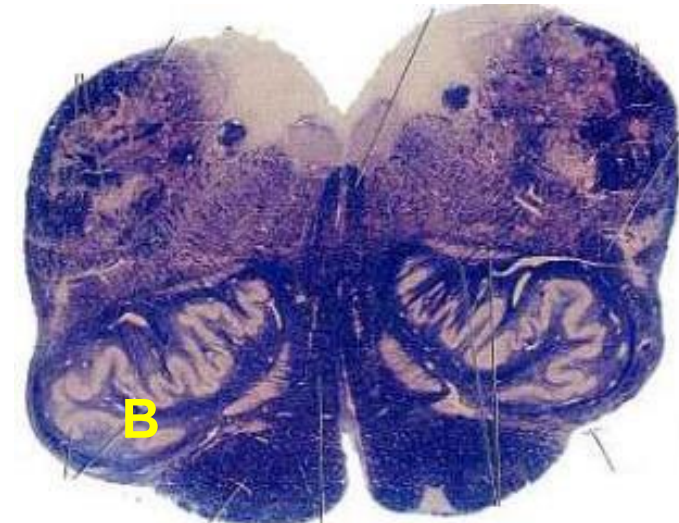
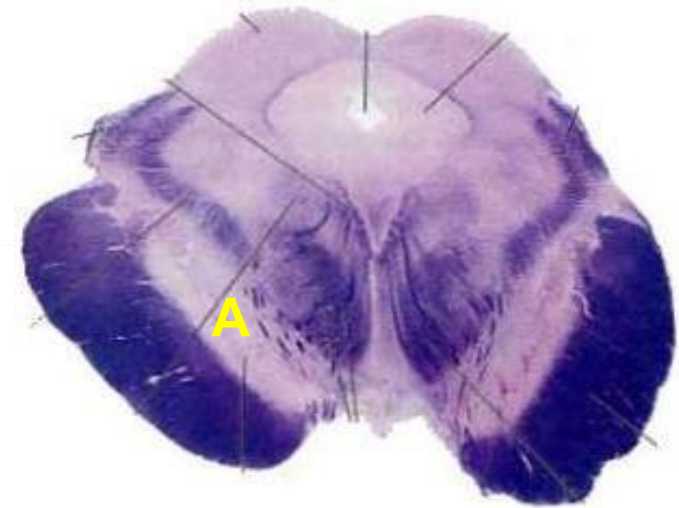
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Name the nuclei!

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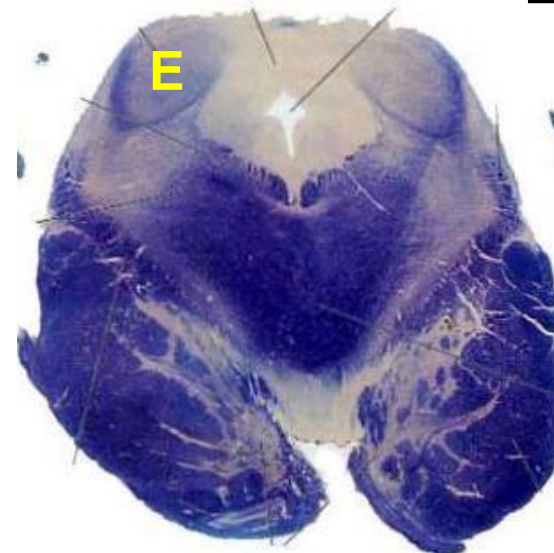
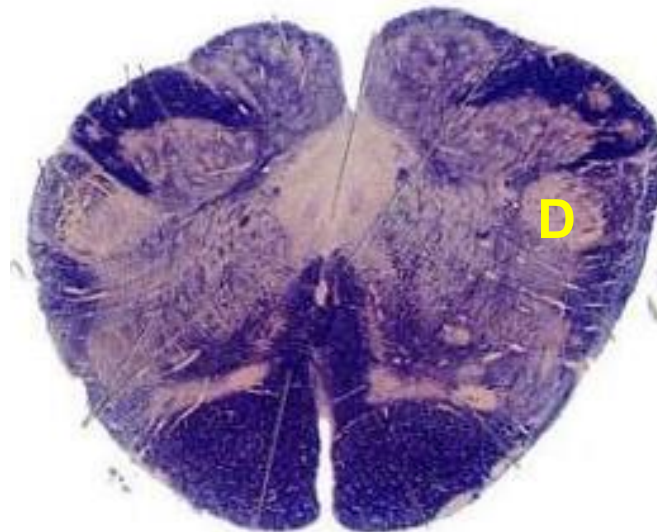
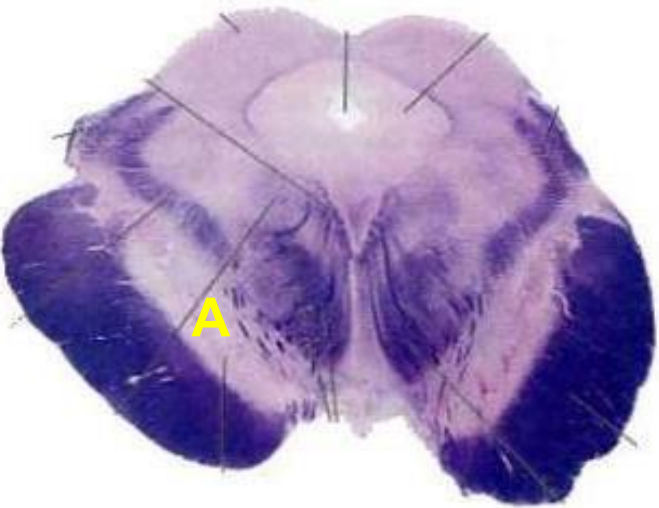
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Name the nuclei!

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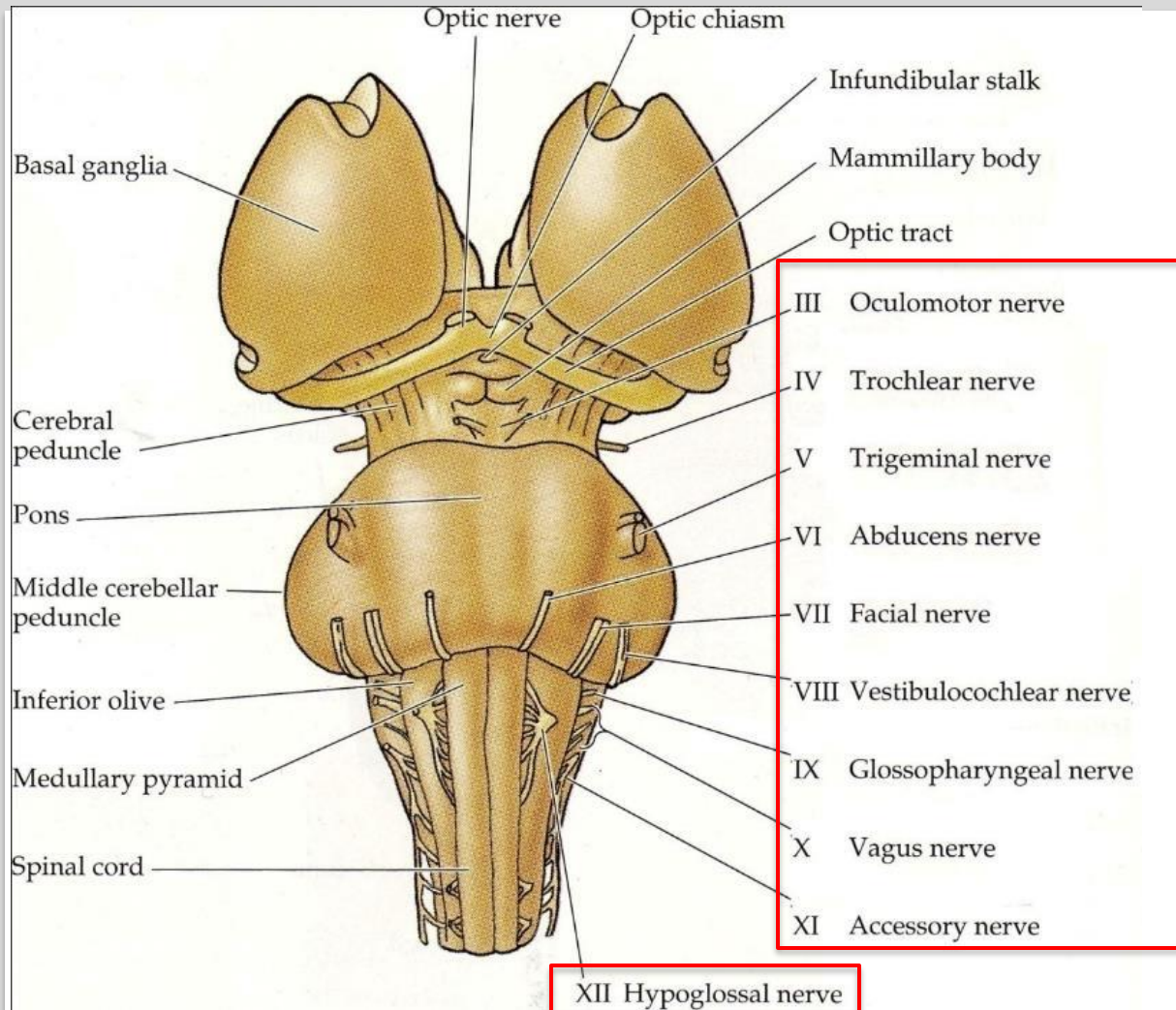
- A – substantia nigra
- B – inferior olive
- C – pontine nuclei
- D – spinal trig. nucleus
- E – inferior colliculus

Cranial nerves

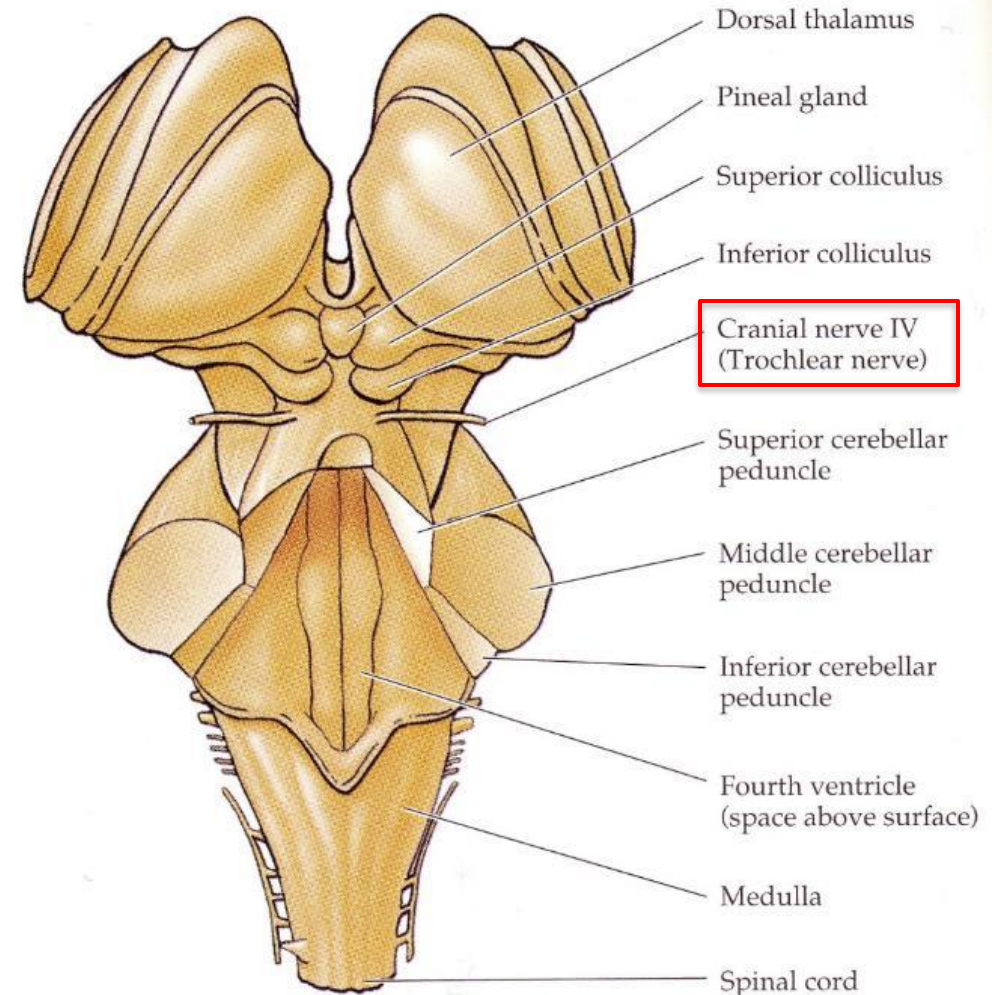
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Ventral View



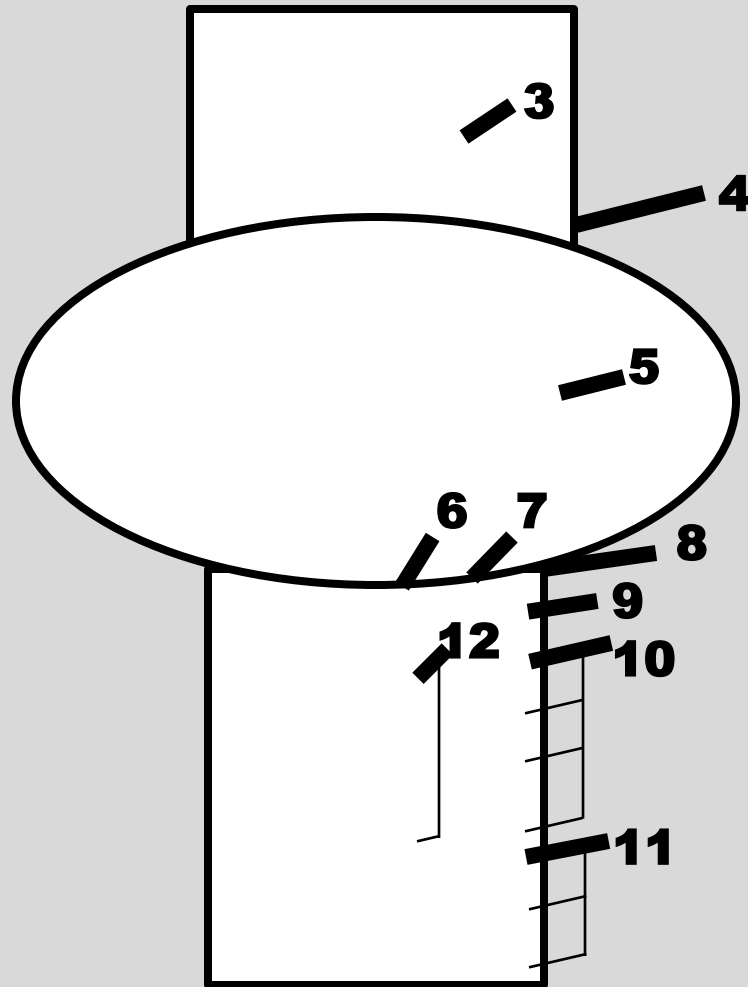
Dorsal View



Cartoon of cranial nerve arrangement

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3 oculomotor

4 trochlear

5 trigeminal

6 abducens

7 facial

8 vestibulocochlear

9 glossopharyngeal

10 vagus

11 spinal accessory

12 hypoglossal

ventral

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Nerves are part of the PNS.
Tracts are part of the CNS.

All cranial nerves innervate
ipsilateral tissue, whether they are
afferents, efferents, or both.

Ipsilateral. Ipsilateral. Ipsilateral.

This is not a hard rule for tracts.
They can convey ipsilateral
information, contralateral
information, or both.

CN 3 →

CN 5 →

CN 6 →

CN 7 →

CN 8 →

CN 9 →

CN 10 →

CN 12 →

CN 11 →

Summary

- The brainstem has distinctive external features for the medulla, pons, and midbrain.
- The tectum is in the dorsal midbrain only. Tegmentum is further ventral and at all divisions. And the basilar division is most anterior/ventral and present at all levels.
- Distinct nuclei are present at each level of the brainstem.
- There is a rational order for numbering the cranial nerves.

Feedback is appreciated!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

